

CHITTAGONG SHAH AMANAT INTL, Bangladesh

WMO: 419780

Lat: 22.250N

Lon: 91.813E

Elev: 4

StdP: 101.28

Time Zone: 6.00 (E06)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 14.1	(c) 15.2	(d) 8.1	(e) 6.7	(f) 24.1	(g) 10.0	(h) 7.6	(i) 23.0	(j) 8.0	(k) 24.1	(l) 7.1	(m) 24.4	(n) 0.9	(o) 20	(p) 0.333

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	6.1	33.9	27.4	33.1	27.2	32.8	27.1	28.8	32.1	28.4	31.5	28.0	31.0	4.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.1	24.3	30.9	27.5	23.4	30.5	27.1	22.9	30.3	94.4	32.0	92.0	31.8	90.4	31.5	34.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.2	(b) 7.0	(c) 6.1	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 8.2	(b) 7.0	(c) 6.1	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.4	20.4	23.0	26.3	28.3	28.9	28.9	28.4	28.5	28.7	28.0	25.4	21.8
	DBStd	3.27	1.65	1.70	1.39	1.44	1.56	1.43	1.19	1.11	1.04	1.43	1.71	2.04
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	5	3	0	0	0	0	0	0	0	0	0	0	2
	CDD10.0	5982	323	364	506	549	585	567	569	574	559	557	462	366
	CDD18.3	2946	68	131	248	299	327	317	311	316	309	299	212	109
	CDH23.3	31238	289	819	2322	3566	4068	3854	3545	3671	3642	3261	1682	520
	CDH26.7	11179	19	152	717	1439	1821	1569	1190	1288	1336	1145	441	61
Wind	WSAvg	2.7	2.3	2.4	2.7	3.6	3.3	3.4	3.7	3.2	2.5	1.7	1.7	1.7
Precipitation	PrecAvg	2946	6	23	63	105	354	637	671	491	315	243	43	10
	PrecMax	3946	53	110	226	287	766	1276	1420	1147	730	618	299	85
	PrecMin	2250	0	0	1	1	108	201	286	156	142	64	0	0
	PrecStd	502	12	33	63	80	181	275	282	215	153	134	64	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.9	31.0	33.2	34.6	34.7	34.2	33.2	33.4	33.9	33.5	32.1	29.2
		MCWB	19.9	19.2	21.3	26.4	28.2	28.2	27.8	27.6	27.7	27.0	24.5	20.9
	2%	DB	27.1	29.6	31.9	33.0	33.8	33.3	32.2	32.8	32.9	32.8	31.0	28.1
		MCWB	18.7	19.8	22.6	26.7	27.8	28.1	27.4	27.5	27.4	26.5	23.9	20.4
	5%	DB	26.1	28.5	30.9	32.1	33.0	32.8	31.4	31.9	32.0	32.0	30.1	27.1
		MCWB	18.1	19.8	23.2	26.2	27.4	27.9	27.2	27.2	27.0	26.2	23.4	20.2
	10%	DB	25.1	27.7	30.0	31.2	32.1	32.0	30.9	31.0	31.1	31.1	29.1	26.2
		MCWB	18.0	19.8	23.2	26.0	27.0	27.5	27.0	27.0	26.8	26.0	22.6	19.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.0	24.2	26.7	28.6	29.4	29.6	29.2	28.7	28.7	28.2	26.4	23.7
		MCDB	25.4	27.0	29.8	32.3	33.1	32.4	31.5	31.8	31.7	31.6	29.5	25.8
	2%	WB	20.8	23.3	26.2	27.8	28.7	28.9	28.2	28.1	28.1	27.6	25.6	22.2
		MCDB	24.1	26.0	29.2	31.1	32.2	32.0	30.7	30.8	31.1	30.9	28.7	25.9
	5%	WB	19.8	22.5	25.6	27.3	28.1	28.4	27.7	27.6	27.6	27.1	25.0	21.4
		MCDB	23.7	25.4	28.6	30.5	31.4	31.5	30.2	30.3	30.6	30.3	28.2	25.2
	10%	WB	19.1	21.8	25.1	26.8	27.7	28.0	27.2	27.2	27.1	26.6	24.4	20.8
		MCDB	23.5	25.0	28.0	29.9	30.8	31.0	29.7	29.8	30.0	29.6	27.5	24.6
Mean Daily Temperature Range	5% DB	MDBR	9.7	9.5	7.9	6.4	6.1	5.2	4.3	4.6	4.9	5.8	7.7	8.6
		MCDBR	10.7	10.4	9.1	7.1	7.0	6.5	5.5	6.0	6.3	6.9	8.0	9.3
		MCWBR	4.6	4.2	4.2	3.5	3.7	3.0	2.6	2.6	2.7	2.8	3.2	3.5
	5% WB	MCDBR	9.9	8.1	7.1	6.3	6.2	6.0	4.9	5.2	5.6	6.0	6.7	8.3
		MCWBR	4.8	4.3	3.9	3.3	3.6	3.2	2.8	2.6	2.7	2.8	3.2	3.8
Clear-Sky Solar Irradiance	taub	0.516	0.516	0.589	0.618	0.638	0.592	0.513	0.495	0.492	0.494	0.496	0.504	
	taud	1.906	1.950	1.831	1.830	1.831	1.957	2.199	2.260	2.250	2.193	2.072	1.973	
	Ebn at Noon	737	774	737	724	705	732	793	810	808	787	755	732	
	Edh at Noon	178	181	212	215	214	187	147	138	138	140	150	161	
All-Sky Solar Radiation	RadAvg	4.49	5.28	5.84	5.93	5.34	4.40	4.41	4.64	4.65	4.60	4.57	4.19	
	RadStd	0.21	0.26	0.34	0.46	0.58	0.48	0.49	0.35	0.27	0.23	0.29	0.24	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page